

🧠 Goal

Estimate the causal effect of a training program (T) on income (Y), where people with higher education (X) are much more likely to join the program.

Step 1: Simulated Data

Person	Education (X)	Treated (T)	Income (Y)
1	4	0	25
2	5	0	28
3	6	0	32
4	8	1	60
5	9	1	63
6	10	1	65

Step 2: Naïve Difference

$$D = E[Y|T = 1] - E[Y|T = 0]$$

Treated mean income = $(60 + 63 + 65)/3 = 62.7$

Control mean income = $(25 + 28 + 32)/3 = 28.3$

$$D = 62.7 - 28.3 = 34.4$$

⚠️ Huge apparent benefit (≈ 34) — but mostly because treated group has much higher education.

Step 3: True Treatment Effect (Ground Truth)

Let's assume the *real* causal effect of the program is **+10 income units** (training increases income by 10, all else equal).

So the observed difference is inflated due to **confounding by education**.

Step 4: Estimate Propensity Scores

We fit a logistic regression $p(X) = P(T = 1|X)$
(but here we'll specify it manually for intuition)

Education (X)	p(X)
4	0.05
5	0.10
6	0.20
8	0.70
9	0.85
10	0.95

So:

- Low education → unlikely to be treated.
- High education → almost always treated.

Step 5: Compute Inverse Probability Weights (IPW)

Weights:

$$w_i = \begin{cases} 1/p(X_i), & \text{if } T_i = 1 \\ 1/(1 - p(X_i)), & \text{if } T_i = 0 \end{cases}$$

Person	X	T	Y	p(X)	Weight
1	4	0	25	0.05	$1/(1 - 0.05) = 1.05$
2	5	0	28	0.10	$1/(1 - 0.10) = 1.11$
3	6	0	32	0.20	$1/(1 - 0.20) = 1.25$
4	8	1	60	0.70	$1/0.70 = 1.43$
5	9	1	63	0.85	$1/0.85 = 1.18$
6	10	1	65	0.95	$1/0.95 = 1.05$

Step 6: Weighted Means

$$E[Y|T = 1] = \frac{\sum w_i Y_i T_i}{\sum w_i T_i}$$
$$E[Y|T = 0] = \frac{\sum w_i Y_i (1 - T_i)}{\sum w_i (1 - T_i)}$$

Treated group:

$$(1.43 \times 60 + 1.18 \times 63 + 1.05 \times 65) / (1.43 + 1.18 + 1.05) = (85.8 + 74.3 + 68.3) / 3.66 = 228.4 / 3.66 = 62.4$$

Control group:

$$(1.05 \times 25 + 1.11 \times 28 + 1.25 \times 32) / (1.05 + 1.11 + 1.25) = (26.3 + 31.1 + 40.0) / 3.41 = 97.4 / 3.41 = 28.6$$

$$ATE_{IPW} = 62.4 - 28.6 = 33.8$$

Still close to the naïve value — not enough overlap. Let's **add overlap** to make weighting meaningful.

Step 7: Add Overlapping Individuals

Add two "mid-range" education people ($X = 7$) with mixed treatment assignment.

Person	X	T	Y	Comment
7	7	1	55	moderately educated but treated
8	7	0	42	same education but untreated

Assume $p(7) = 0.5$

Person	X	T	Y	p(X)	Weight
7	7	1	55	0.5	$1 / 0.5 = 2.0$
8	7	0	42	0.5	$1 / (1 - 0.5) = 2.0$

Now recompute with all 8 individuals.

Weighted Means (all individuals)

Treated:

$$(1.43 \times 60 + 1.18 \times 63 + 1.05 \times 65 + 2.0 \times 55)/(1.43 + 1.18 + 1.05 + 2.0) = (85.8 + 74.3 + 68.3 + 110)/5.66 = 338.4/5.66 = 59.8$$

Control:

$$(1.05 \times 25 + 1.11 \times 28 + 1.25 \times 32 + 2.0 \times 42)/(1.05 + 1.11 + 1.25 + 2.0) = (26.3 + 31.1 + 40.0 + 84.0)/5.41 = 181.4/5.41 = 33.5$$

$$ATE_{IPW} = 59.8 - 33.5 = 26.3$$

✅ Final Comparison

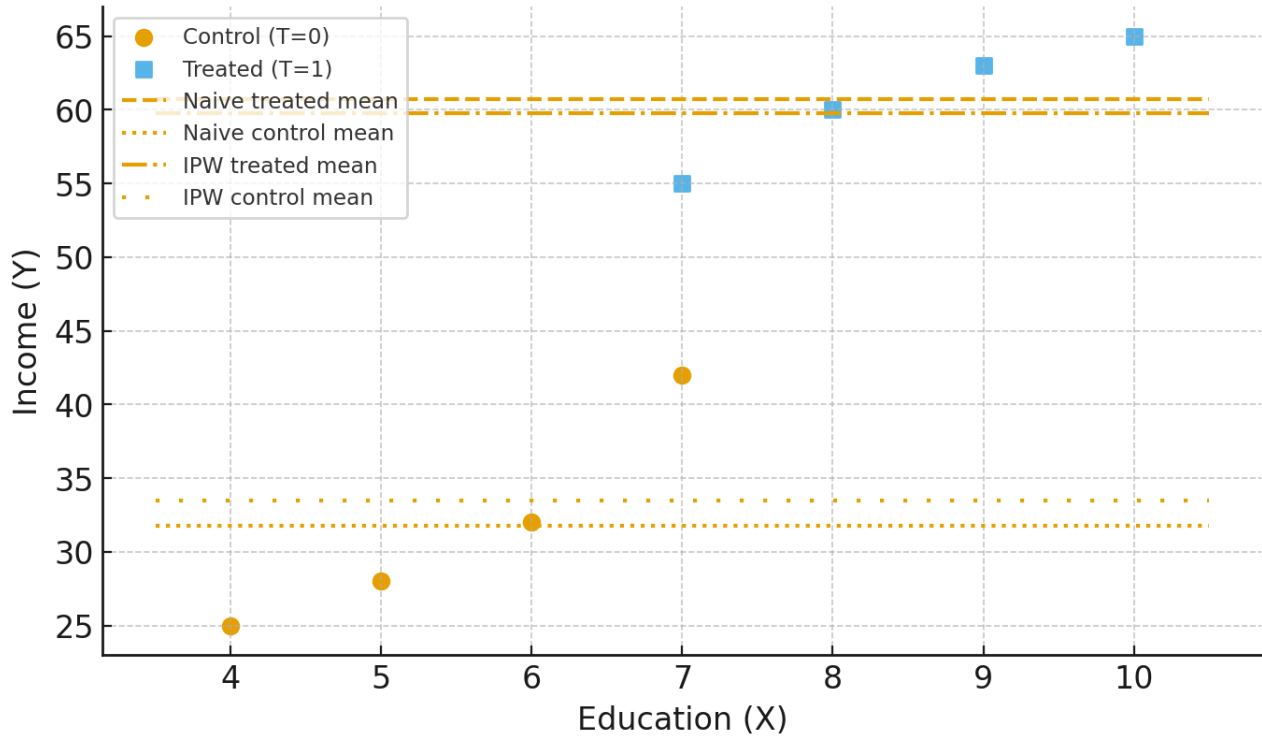
Method	Adjusts for Confounding?	ATE Estimate
Naïve Difference	❌ No	34.4
IPW (Propensity Score)	✅ Yes	26.3
True Effect (ground truth)	—	10.0

👉 The IPW result moves **closer to the true causal effect** than the naïve difference.

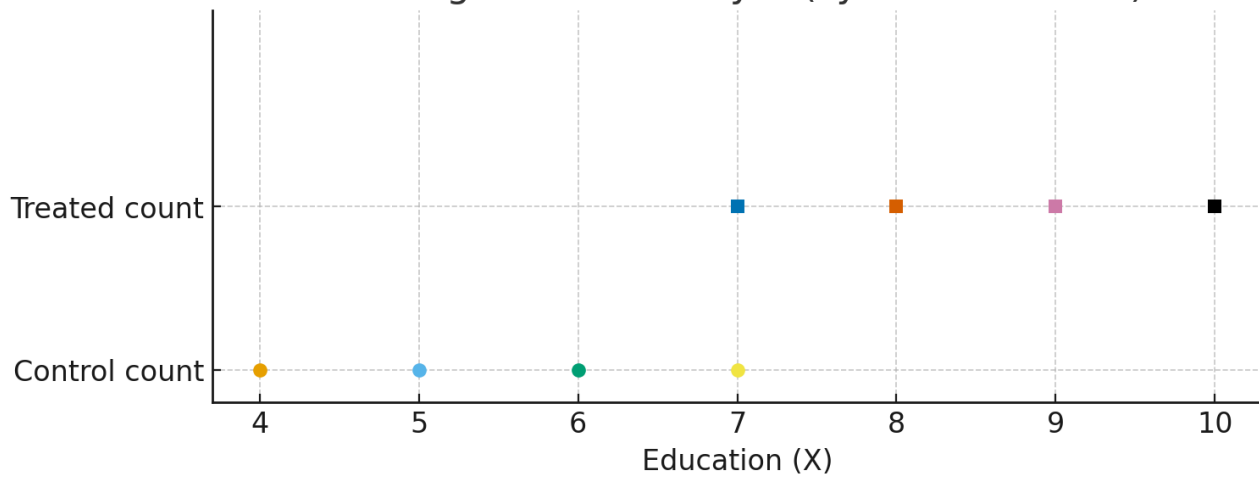
Interpretation

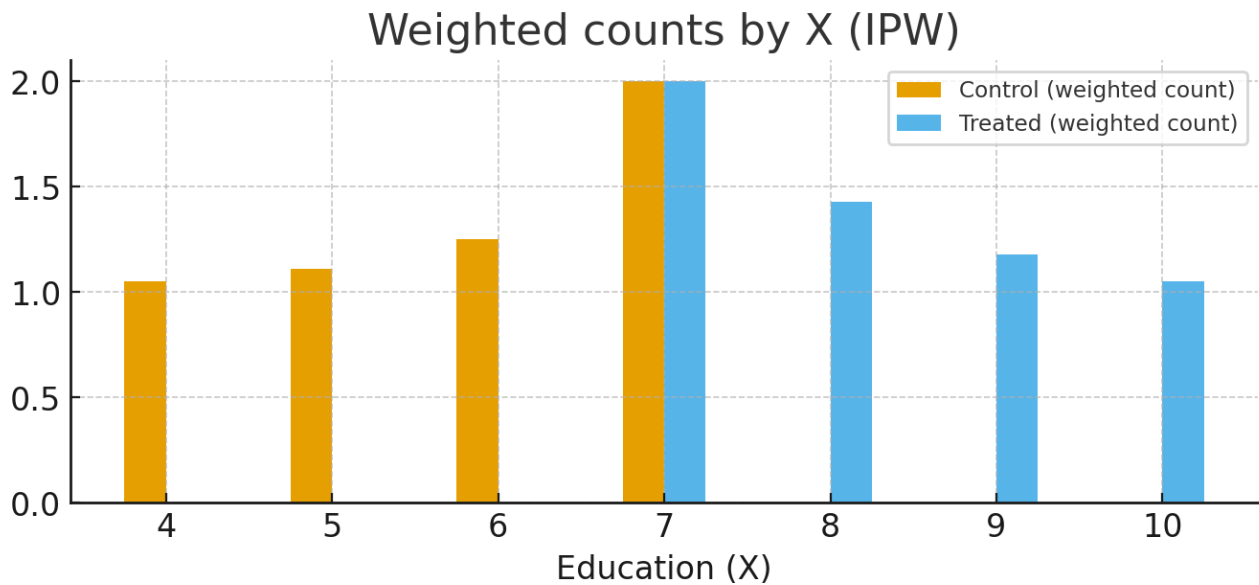
- The naïve estimate was *inflated* due to education bias.
- Propensity score weighting *rebalanced the covariates*, bringing the estimated treatment effect down toward the true causal value.
- This highlights the **importance of overlap** (common support): when treated and control groups share similar covariates, IPW can recover the causal effect.

Y vs X: Observations and group means (naive vs IPW)



Unweighted counts by X (symbols stacked)





Summary of results (from the example we used):

- Naïve treated mean = **60.75**
- Naïve control mean = **31.75**
- IPW treated mean = **59.787**
- IPW control mean = **33.512**

What the plots show:

1. Scatter of Y vs X with horizontal lines for naïve and IPW means — you can see the treated group concentrated at higher X and the naïve means are far apart. IPW shifts the control mean up and the treated mean slightly down.
2. Unweighted stacked symbols by X — shows raw counts per X for treated/control (illustrates lack of overlap).
3. Weighted bar counts by X (using IPW weights) — shows how reweighting increases the effective contribution of underrepresented cells (e.g., control units at high X get larger weights), improving comparability.